1. **Create a simple JavaScript function called calculateTotal that takes two numbers: itemPrice and quantity, and returns the total bill amount using arithmetic operators.**

```
function calculateTotal(itemPrice, quantity) {
    return itemPrice * quantity;
}

// Example
let itemPrice = 250;
let quantity = 3;

let totalBill = calculateTotal(itemPrice, quantity);

console.log("Total Bill Amount: ₹" + totalBill);
```

2. Build a Flipkart-style discount calculator: given product price, discount percentage, and a boolean isMember, use arithmetic and logical operators to calculate the final price (apply an extra 5% off if isMember is true).

```
function calculateFinalPrice(productPrice, discountPercentage, isMember) {
    // Apply the regular discount
    let finalPrice = productPrice - (productPrice * discountPercentage / 100);

    // Apply an extra 5% discount for members
    if (isMember) {
        finalPrice = finalPrice - (finalPrice * 5 / 100);
    }

    return finalPrice;
}

// Example
let productPrice = 2000;
let discountPercentage = 20;
let isMember = true;

let finalPrice = calculateFinalPrice(productPrice, discountPercentage,
isMember);

console.log("Original Price: ₹" + productPrice);
console.log("Discount: " + discountPercentage + "%");
console.log("Member: " + isMember);
console.log("Final Price: ₹" + finalPrice.toFixed(2));
```

**3. Write a function isEligibleForOffer that takes a user's age and total order value, and returns true if the user is 18 or older AND the order value is above 500, otherwise false.

Hint: Use relational and logical operators together.**

```
function isEligibleForOffer(age, totalOrderValue) {
    return age >= 18 && totalOrderValue > 500;
}

// Example 1
console.log(isEligibleForOffer(20, 750)); // true

// Example 2
console.log(isEligibleForOffer(16, 800)); // false

// Example 3
console.log(isEligibleForOffer(25, 450)); // false
```

4. Given three variables: likes, comments, and shares (all numbers), write code to check if a post is 'trending' on Instagram (at least 1000 likes OR more than 200 comments AND at least 50 shares). Print the result.

```
// Variables
let likes = 850;
let comments = 250;
let shares = 60;

// Check if the post is trending
let isTrending = (likes >= 1000) || (comments > 200 && shares >= 50);

// Print the result
console.log("Trending:", isTrending);
```

5. Write a code snippet that demonstrates the difference between pre-increment (++count) and post-increment (count++) by logging the values before and after using both on a followerCount variable.

```
// Pre-increment (++count)
let followerCount = 100;

console.log("Initial follower count:", followerCount);
console.log("Using pre-increment (++followerCount):", ++followerCount);
console.log("Follower count after pre-increment:", followerCount);

// Post-increment (count++)
followerCount = 100; // Reset value

console.log("\nInitial follower count:", followerCount);
console.log("Using post-increment (followerCount++):", followerCount++);
console.log("Follower count after post-increment:", followerCount);
```